

appears across the given 6.0 Ω resistor.

25	B	Option	p.d. across both resistors	p.d. across top resistor	Potential of point J
		A	$8 - (-8) = 16 \text{ V}$	$\frac{2}{2+2} \times 16 = 8 \text{ V}$	$8 - V_J = 8$ $V_J = 0 \text{ V}$
		B	16 V	$\frac{6}{6+2} \times 16 = 12 \text{ V}$	$8 - V_J = 12$ $V_J = -4 \text{ V}$ (as required)
		C	16 V	$\frac{2}{2+6} \times 16 = 4 \text{ V}$	$8 - V_J = 4$ $V_J = 4 \text{ V}$
		D	16 V	$\frac{6}{6+8} \times 16 = 6.86 \text{ V}$	$8 - V_J = 6.86$ $V_J = 1.14 \text{ V}$
26	C	The force is directly proportional to the product of the currents			